

Algonquin College

Faculty of Technology and Trades
School of Advanced Technology

Main Assignment - Part 1

Build a Business Case
CanTech Data Centre - Winnipeg, Manitoba

Course: CST8809 - Business Continuity & Disaster Recovery

Section: 26S_CST8809_010

Submitted by: Mohammad Zghoul

Submitted to: Professor Sazzad Hossain

Date of submission: May 22, 2026

Spring 2026

Section i - Stakeholder Matrix

The matrix below identifies ten potential stakeholders in CanTech's Business Continuity / Disaster Recovery program. Stakeholders marked with an "X" are considered key. The five highest-priority stakeholders are numbered 1-5 in the order they should be engaged.

Stakeholder	Key (X)	Priority (Top 5)
Chief Operations Officer (COO)	X	1
Chief Executive Officer (CEO)	X	2
Chief Financial Officer (CFO)	X	3
Infrastructure Manager	X	4
Information Systems (IS) Division Manager	X	5
External Auditors	X	
Major Customers (oil & gas, healthcare, financial, government)	X	
Board of Directors / Shareholders	X	
Network and Systems Admin		
Human Resources / Corporate Safety Officer		

Section ii - Top 5 Stakeholder Justifications

1. Chief Operations Officer (COO)

The COO is the BCP Manager's direct supervisor and the executive sponsor of the business continuity program. They control day-to-day prioritization, headcount, and inter-departmental access, all of which are essential to a single-FTE BCP project with an 8-12 month deadline. A supportive COO unlocks the IS Division and the Infrastructure team, secures meeting time with the CEO and CFO, and signs off on policy and scope. Conversely, if the COO does not internalize the audit findings or treats BCP as IS-Division-only, the project will stall, scope creep will go unchecked, and contractual SLA exposure will continue to grow. Engaging the COO first is therefore the single highest-leverage action in the program.

2. Chief Executive Officer (CEO)

The CEO sets corporate priority and, per the scenario, was personally involved in selecting the BCP Manager. Their endorsement gives the program organizational "air cover" across Winnipeg, Toronto, and Vancouver and signals to the board and external auditors that CanTech is responding to the audit findings. A well-briefed CEO will defend budget allocations and resist competing strategic initiatives that crowd out BCP work. A poorly-briefed CEO is the largest negative risk: with two co-owners (Ronstadd and Branson) and a different mental model of what BCP entails for each executive, conflicting direction can derail the project mid-flight. Engaging the CEO with a clear, scope-bounded mandate is therefore the second priority.

3. Chief Financial Officer (CFO)

The CFO controls the capital required for cooling, redundancy, hotsite testing, cybersecurity tooling, and business-interruption insurance — none of which are presently funded. With Year-2 revenue projected at \$100M and worst-case single-event exposure exceeding \$13M (untested DR failure), the CFO's risk-versus-spend lens is critical to a defensible business case. A CFO who accepts the financial modelling will release multi-year funding. A CFO who pushes back will force the program into incrementalism, leaving the largest exposures un-mitigated. Because CFO approvals gate every other workstream, the CFO must be engaged before detailed proposals are tabled.

4. Infrastructure Manager

The Infrastructure Manager owns the hardware, software, security systems, and physical plant — the entire technical surface area the BCP must protect. They will own the day-to-day mitigation work (cooling redundancy, change management, asset inventory) and are the most credible internal source of effort, cost, and feasibility data.

A cooperative Infrastructure Manager accelerates the BIA, supplies accurate dependency maps, and validates Recovery Time Objectives. A disengaged one creates the risk of inaccurate or self-serving estimates and territorial resistance, particularly because BCP touches IS Division turf. They must be engaged early as a partner rather than as a subject of audit.

5. Information Systems (IS) Division Manager

The IS Division Manager currently owns the only existing DR effort — the 5-year hotsite contract that the audit found unreliable. They are the most affected stakeholder by the new program and, depending on how the BCP project is framed, the most likely source of resistance. Positively, they hold all the institutional knowledge about historical DR decisions, customer technical commitments, and the previous purchased plan. Negatively, the BCP project will surface uncomfortable findings (untested recovery, mismatched hardware, stale team rosters). Engaging the IS Division Manager as a co-author rather than a target is essential to converting them into an ally and accelerating remediation.

Section iii - Top 5 IT Risks and Financial Impact

Financial-impact estimates use Year-2 sales projection of \$100,000,000 (CanTech scenario, Sales Forecast). Hourly revenue baseline = \$100M / 8,760 hrs ≈ \$11,416/hr. All figures are CAD.

Potential Risk (IT)	Estimated Financial Impact (with Calculation)
Extended power outage (>72 hrs) at the Winnipeg site	Per-incident SLA limit = 72 hrs. Hourly revenue baseline = \$100M (Year-2 projection) / 8,760 hrs ≈ \$11,416/hr. A 72-hour outage = 72 × \$11,416 ≈ \$821,952 direct revenue lost. Adding contractual service credits (~2× hourly rate) and emergency generator-fuel/labour adds ~\$0.8M, for a total exposure of ≈ \$1.6M per incident.
Cyberattack (ransomware / DDoS / data breach)	Audit explicitly notes "no provision for cybersecurity has been addressed." Industry-average Canadian ransomware incident = CAD \$5.7M (IBM Cost of a Data Breach 2024). Multi-tenant client base (oil & gas, healthcare, finance) raises class-action exposure. Conservative single-event exposure: \$3M ransom/recovery + \$2M client lawsuits + \$1M brand damage ≈ \$6M.
Untested DR / hotsite vendor failure during invocation	Existing 5-yr contract is in Year 4, untested, and no longer matches current hardware. Worst case: site lost 2 weeks (336 hrs) and hotsite cannot accept the workload. 336 × \$11,416 ≈ \$3.84M direct revenue + 50% Y2 churn on enterprise accounts (assume 10% of \$100M revenue) = \$10M customer-attribution write-off → total ≈ \$13.8M.
Customer SLA breach (>48-72 hr per incident or >700 hrs/yr)	Contract caps annual outage at 700 hrs. Exceeding 700 hrs typically triggers full SLA-credit refunds equal to ~5% of contract value. On the \$100M Year-2 revenue base, 5% = \$5M of credits/refunds, before any litigation. Likely cascades to ~\$5-7M event.
Mainframe / storage hardware failure (Winnipeg, Toronto, or Vancouver)	Mainframes run at 70-80% capacity with no spare headroom. Audit states maximum user tolerance without mainframe = 5 business days (120 hrs). 120 × \$11,416 = \$1.37M direct revenue loss + 72-hr replacement-equipment commitment to customers requiring \$200K of expedited hardware shipments = ~\$1.6M per event.

Section iv - Top 5 Risk Justifications and Mitigations

1. Extended power outage (>72 hrs)

Manitoba Hydro outages of 24-72 hrs are credible during severe winter storms, ice events, or grid failures. CanTech budgeted only \$3M for generation and back-ups, and the audit notes the DR plan has never been tested, so generator endurance, fuel resupply, and automated transfer behaviour are unproven. A 72-hour outage immediately breaches the 48-72 hr customer SLA, triggers service credits, and risks customer churn. The financial exposure (~\$1.6M per event) is meaningful relative to a \$100M revenue base. Mitigation: dual feeds where available, N+1 generators with multi-day fuel contracts, periodic load-bank tests, and a formal black-start drill at least annually.

2. Cyberattack (ransomware / DDoS / data breach)

This is the highest residual risk because the audit confirms there is no cybersecurity program. CanTech is internet-exposed (customer portals, voice-response units, mobile sales force connectivity) and hosts client data for high-value sectors. A single ransomware event could halt operations across all three branches and trigger class-action exposure on top of recovery cost. Average Canadian incident cost $\approx$ CAD \$5.7M. Mitigation: stand up a baseline control set (EDR, SIEM, MFA, hardened backups with offline copies, perimeter segmentation, incident-response runbooks, mandatory security awareness training) and align to NIST CSF or ISO 27001 within 12 months.

3. Untested DR / hotsite vendor failure

The 5-year hotsite contract is in Year 4, has never been tested, has not been updated since signing, and the production hardware behind it has changed several times. If invoked, the probability the vendor cannot accept the workload is high, turning a recoverable incident into an existential one. Worst-case exposure $\approx$ \$13.8M including customer attrition. Mitigation: schedule a tabletop within 60 days, a partial failover within 6 months, and a full failover annually; amend the contract to reflect current hardware and bandwidth requirements; verify off-site backup completeness before each test.

4. Customer SLA breach (>48-72 hrs per incident or >700 hrs/yr)

The SLA structure is binding and quantitative: any breach immediately produces service-credit liability, and accumulated breaches risk contract loss and litigation. Because CanTech has no Business Interruption insurance, every dollar of credit lands on the income statement. With Year-2 revenue at \$100M and a 5% credit assumption, a single bad year of breaches is \$5M+ in direct cost. Mitigation: instrument SLA dashboards per customer, raise an early-warning alert at 70% of the annual cap, redesign weakest single-points-of-failure first, and negotiate tiered credit structures on renewal.

5. Mainframe / storage hardware failure

The three regional mainframes (Winnipeg, Toronto, Vancouver) run at 70-80% capacity with no spare headroom; the audit further notes the contract with the recovery vendor lists hardware that no longer matches production. A failure of any one mainframe cascades across regional and sales offices because order entry and customer gateways depend on these systems. Direct revenue exposure $\approx$ \$1.4-1.6M per event. Mitigation: build a documented spare-parts depot, validate clustered/active-active configurations or warm-standby capacity at peer sites, and verify backup-restore integrity for each mainframe quarterly.

Section v - Recommended CanTech IT Improvements

Recommended Improvement	Benefit (25-50 words)	Qualitative & Quantitative Support (25-50 words)
Annual test of the DR / hotsite contract (full failover + tabletop)	Closes the audit finding that the existing 5-year hotsite contract has never been tested. Confirms actual Recovery Time Objective and Recovery Point Objective and surfaces gaps before a real incident.	Qualitative: removes the single largest open audit issue and demonstrates due diligence to shareholders. Quantitative: estimated \$50-100K/yr in testing fees vs. \$13.8M worst-case DR-failure exposure (Risk 3) — payback in well under one event.
Establish a formal cybersecurity program (controls, monitoring, IR plan)	Replaces the current state of "no provision for cybersecurity" with a defensible control set (perimeter, EDR, SIEM, IAM, hardening, incident response, awareness).	Qualitative: protects multi-sector client data and meets implicit customer/regulatory expectations. Quantitative: program cost $\sim$ \$0.5-1M/yr vs. \$6M average single-event ransomware exposure (Risk 2); also unlocks insurability.
Centralized IT asset inventory and change-management process	Provides accurate counts of workstations, software, and hardware acquisitions and prevents change-induced outages by routing all	Qualitative: addresses the audit finding of "no accurate inventory of workstations, no accounting for hardware/software acquisition."

Recommended Improvement	Benefit (25-50 words)	Qualitative & Quantitative Support (25-50 words)
	production changes through a Change Advisory Board.	Quantitative: industry data shows formal change management reduces change-related incidents by 30-50%, protecting \$1-3M/yr in avoided downtime.
Active geographic redundancy across Winnipeg, Toronto, and Vancouver	Removes Winnipeg as a single point of failure for the 10,000+ servers. Workload can fail over to Toronto/Vancouver, sustaining the 48-72 hr customer SLA even during a regional disaster.	Qualitative: directly mitigates flood, fire, and prolonged power risks unique to Winnipeg. Quantitative: incremental cross-site replication and capacity uplift ~\$1.5-2M one-time + ~\$0.5M/yr — protects the \$5-7M annual SLA-credit exposure (Risk 4).
Procure business-interruption, contingent-BI, and extra-expense insurance	Closes the explicit audit gap that CanTech carries no BI, contingent-BI, or extra-expense coverage, transferring residual financial risk to an insurer once mitigations are in place.	Qualitative: protects directors and shareholders from personal-liability exposure called out in the audit. Quantitative: typical BI premium \$150-300K/yr on a \$100M revenue base, vs. multi-million-dollar single-event exposures across Risks 1-5.

Note. This Business Case focuses on the IT side per the assignment instructions. Subsequent Parts 2 and 3 of the Main Assignment will translate these stakeholders, risks, and improvements into a Business Impact Assessment and a formal BCP/DR Plan.